

Comptia A+ Assignment

[A+ - Understanding and maintenance of Networks]

Configuring Basic Network Hardware

IP addressing basics and network configuration. Set up a router with default settings.

Task

❖ **1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page. If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.**

➤ Accessing the Router Admin Panel:

➤ Objective:

- To access the home Wi-Fi router's administration panel using its Default Gateway IP address and verify the router login page.

➤ Step 1: Connect to the Wi-Fi Network:

- First, I connected my laptop/PC to the home Wi-Fi network whose router I wanted to configure.

➤ What I am checking:

- The computer must be connected to the router before accessing its administration page.

➤ Step 2: Open Command Prompt:

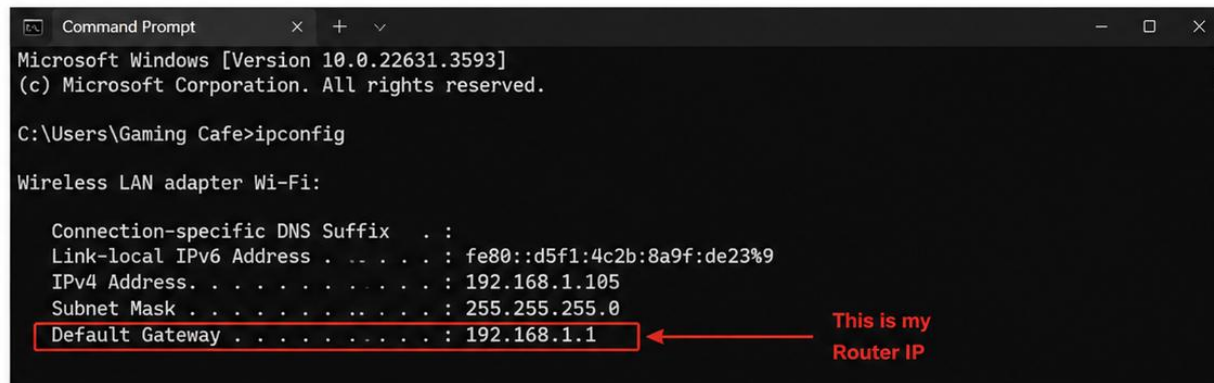
- I opened Command Prompt in Windows by pressing:
- Windows + R → type cmd → Enter
- Then I entered:
- ipconfig

➤ **What I am checking:**

- I looked under the active network adapter for **Default Gateway**.

➤ For example:

- Default Gateway : 192.168.1.1
- The address shown on **my own computer** is the router's local IP address.



```
Microsoft Windows [Version 10.0.22631.3593]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Gaming Cafe>ipconfig

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::d5f1:4c2b:8a9f:de23%9
    IPv4 Address. . . . . : 192.168.1.105
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
```

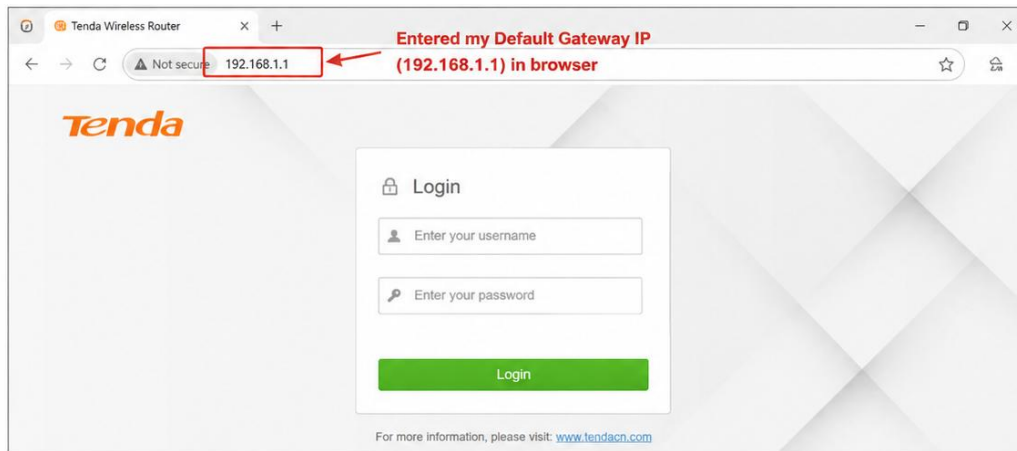
This is my Router IP

➤ Step 3: Open the Router Login Page:

- I opened a web browser and entered the Default Gateway address in the address bar.

➤ For example:

- http://192.168.1.1



➤ Step 4: Verify the Login Page:

- The router's administration/login page appeared. I verified that the page belongs to the router before entering any credentials.

➤ What I am checking:

- Router login page is accessible.
- Router/admin interface is responding.
- Correct Default Gateway address is being used.

❖ **2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.**

➤ Find Network Configuration Details:

- I connected my laptop to the Wi-Fi network and used the Command Prompt to find the network configuration.
- I opened Command Prompt and entered:
- **Ipconfig**

- Under the active Wireless LAN adapter (Wi-Fi) section, I noted the following details:

Network Detail	Value
IPv4 Address	192.168.1.100
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

❖ **3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access. Remember to revert to automatic/DHCP after testing to avoid network issues.**

➤ Static IP Configuration and Internet Connectivity Test:

➤ Objective

- To change the laptop's IP configuration from **Automatic (DHCP)** to a **Static IP address**, test Internet connectivity, and finally restore the network settings to **Automatic (DHCP)**.

➤ Step 1: Check the Current Network Configuration:

- First, I connected the laptop to the Wi-Fi network and opened Command Prompt. I entered:

➤ **ipconfig**

➤ I noted the current **IPv4 Address, Subnet Mask, and Default Gateway**.

```
Command Prompt
Microsoft Windows [Version 10.0.22631.3527]
(c) Microsoft Corporation. All rights reserved.

C:\Users\gamer>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi:

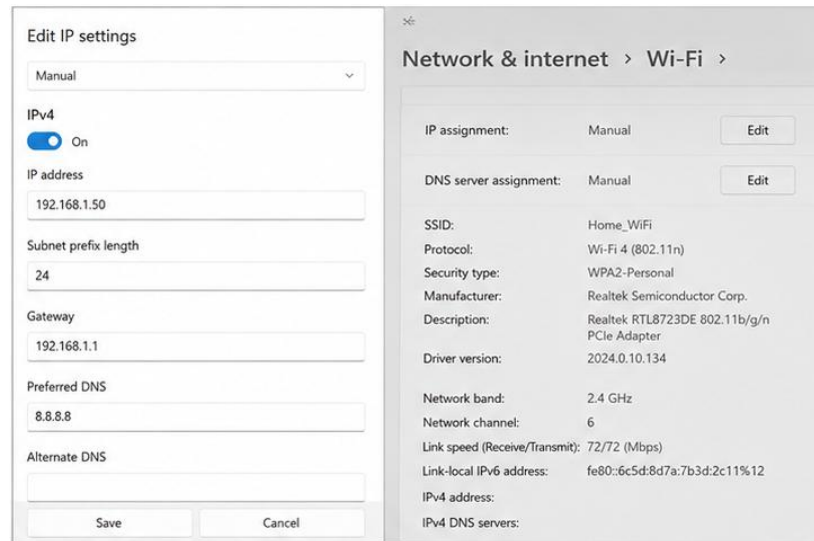
    Connection-specific DNS Suffix . . : 
    Link-local IPv6 Address . . . . . : fe80::6c5d:8d7a:7b3d:2c11%12
    IPv4 Address. . . . . : 192.168.1.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\gamer>
```

➤ Step 2: Change IP Assignment to Manual:

- I opened Windows Network Settings and selected the connected Wi-Fi network. Under IP Assignment, I selected **Edit**, changed the setting from **Automatic (DHCP)** to **Manual**, and enabled **IPv4**.
- I then entered a static IP address in the same network range as my router.
- For example, if the router's address is 192.168.1.1, a suitable example configuration is:

Setting	Example
IP Address	192.168.1.50
Subnet Prefix Length	24
Gateway	192.168.1.1
Preferred DNS	8.8.8.8



➤ Step 3: Test Internet Connectivity:

- After saving the static IP configuration, I tested the Internet connection by opening a website. I also used Command Prompt to verify connectivity:
- **ping 8.8.8.8**
- If the ping returns **Reply from...** messages and websites open normally, the Internet connection is working with the static configuration.

```
Command Prompt
C:\Users\gamer>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=20ms TTL=116
Reply from 8.8.8.8: bytes=32 time=19ms TTL=116
Reply from 8.8.8.8: bytes=32 time=19ms TTL=116
Reply from 8.8.8.8: bytes=32 time=20ms TTL=116

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 19ms, Maximum = 20ms, Average = 19ms

C:\Users\gamer>
```

➤ Step 4: Restore Automatic (DHCP):

- After completing the test, I returned to the Wi-Fi IP settings and changed **IP Assignment** back to **Automatic (DHCP)**. I saved the settings and confirmed that the laptop could connect to the Internet normally.



➤ Result:

- The laptop was successfully configured with a static IP address, and Internet connectivity was tested. After testing, the network configuration was restored to **Automatic (DHCP)** to avoid future network connectivity problems.

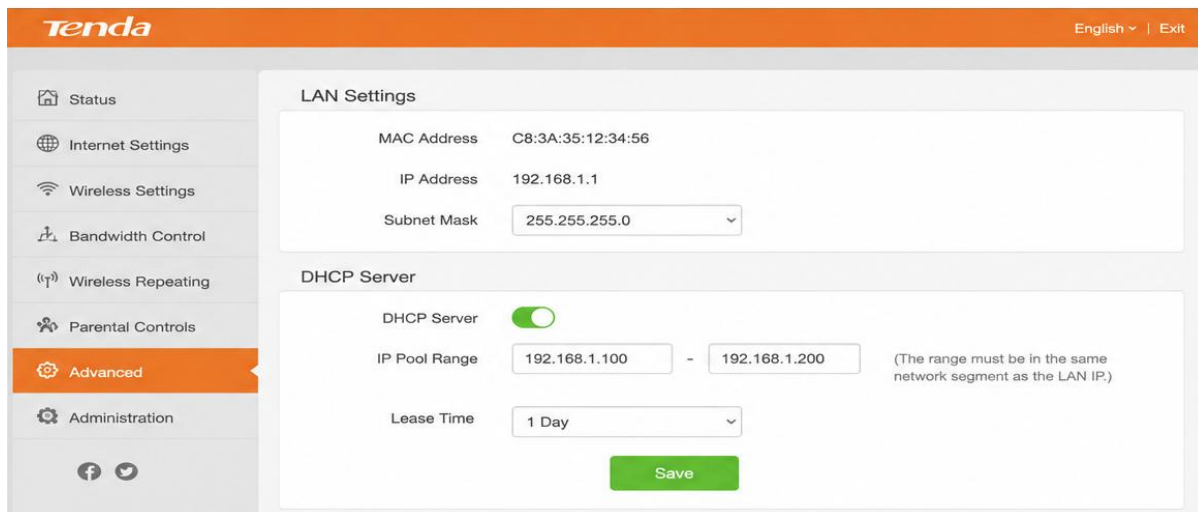
❖ 4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

➤ DHCP Range and IP Conflict:

- I logged in to the router's admin panel and opened the **DHCP Server/LAN settings** section. The current DHCP address range was:
- **DHCP Range: 192.168.1.100 – 192.168.1.200**

➤ What happens if two devices get the same IP address?

- If two devices are assigned the same IP address, an IP address conflict occurs. This can cause network connectivity issues for one or both devices, such as the inability to access the internet or local network resources, because the router cannot distinguish between the two devices. It may also lead to intermitted disconnections.



The screenshot shows the Tenda router's web interface. The left sidebar contains navigation links: Status, Internet Settings, Wireless Settings, Bandwidth Control, Wireless Repeating, Parental Controls, **Advanced** (highlighted), and Administration. The main content area is titled 'LAN Settings' and contains two sections. The 'LAN Settings' section shows the MAC Address as C8:3A:35:12:34:56, the IP Address as 192.168.1.1, and the Subnet Mask as 255.255.255.0. The 'DHCP Server' section shows the DHCP Server toggle switch turned on, the IP Pool Range set to 192.168.1.100 - 192.168.1.200, and the Lease Time set to 1 Day. A green 'Save' button is located at the bottom right of the DHCP Server section. A note next to the IP Pool Range states: '(The range must be in the same network segment as the LAN IP.)'